Example 9.38

Determine if f defined by $f(x) = \begin{cases} x^2 \sin\frac{1}{x}, & \text{if } x \neq 0 \\ 0, & \text{if } x = 0 \end{cases}$ is continuous in $\mathbb{R}$.

Solution

By Sandwitch theorem $\lim_{x\to 0} x^2 \sin\frac{1}{x} = 0$ and $f(0) = 0$ by the definition of $f(x)$. Hence it is continuous at $x = 0$. For other values it is clearly continuous and hence continuous in $\mathbb{R}$.

EXERCISE 9.5

(1) Prove that $f(x) = 2x^2 + 3x - 5$ is continuous at all points in $\mathbb{R}$.

(2) Examine the continuity of the following :

(i) $x + \sin x$ (ii) $x^2 \cos x$ (iii) $e^x \tan x$

(iv) $e^{2x} + x^2$ (v) $x . \ln x$ (vi) $\frac{\sin x}{x^2}$

(vii) $\frac{x^2 - 16}{x + 4}$ (viii) $|x + 2| + |x - 1|$ (ix) $\frac{|x - 2|}{|x + 1|}$

(x) $\cot x + \tan x$

(3) Find the points of discontinuity of the function f, where

(i) $f(x) = \begin{cases} 4x + 5, & \text{if } x \leq 3 \\ 4x - 5, & \text{if } x > 3 \end{cases}$ (ii) $f(x) = \begin{cases} x + 2, & \text{if } x \geq 2 \\ x^2, & \text{if } x < 2 \end{cases}$

(iii) $f(x) = \begin{cases} x^3 - 3, & \text{if } x \leq 2 \\ x^2 + 1, & \text{if } x > 2 \end{cases}$ (iv) $f(x) = \begin{cases} \sin x, & 0 \leq x \leq \frac{\pi}{4} \\ \cos x, & \frac{\pi}{4} < x < \frac{\pi}{2} \end{cases}$

(4) At the given point x_0 discover whether the given function is continuous or discontinuous citing the reasons for your answer :

(i) $x_0 = 1$, $f(x) = \begin{cases} \frac{x^2 - 1}{x - 1}, & x \neq 1 \\ 2, & x = 1 \end{cases}$ (ii) $x_0 = 3$, $f(x) = \begin{cases} \frac{x^2 - 9}{x - 3}, & \text{if } x \neq 3 \\ 5, & \text{if } x = 3 \end{cases}$

(5) Show that the function $\begin{cases} \frac{x^3 - 1}{x - 1}, & \text{if } x \neq 1 \\ 3, & \text{if } x = 1 \end{cases}$ is continuous on $(-\infty, \infty)$

(6) For what value of α is this function $f(x) = \begin{cases} \frac{x^4 - 1}{x - 1}, & \text{if } x \neq 1 \\ \alpha, & \text{if } x = 1 \end{cases}$ continuous at $x = 1$?

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